

MAEPiMS Challenge 2026

Nigeria Influenza Forecasting Challenge

Forecasting Seasonal Influenza Using Simulated National Surveillance Data

Background

Seasonal influenza remains one of the leading causes of respiratory illness worldwide. Although influenza surveillance in Nigeria is evolving, forecasting seasonal influenza activity is essential for preparedness, healthcare planning, and epidemic response.

The MAEPiMS Challenge 2026 provides participants with a simulated (synthetic) national influenza surveillance dataset representing realistic influenza transmission across Nigeria. The dataset represents multi-wave influenza dynamics, taking into consideration Nigeria's demographic, climatic, and healthcare context.

NB: THE DEVELOPED MODEL MUST BE ORIGINAL AND NOT DERIVED FROM EXISTING LITERATURE. PARTICIPANTS ARE ENCOURAGED TO INCORPORATE NEW IDEAS AND INNOVATIVE APPROACHES THAT GO BEYOND ESTABLISHED FRAMEWORKS.

Challenge Objective

Develop an original forecasting model capable of accurately predicting influenza activity across Nigeria using **ONLY** the provided simulated surveillance dataset. Mathematical, statistical, machine learning, deep learning, Bayesian, hybrid, and mechanistic models are all permitted.

Dataset Description

The challenge dataset contains three simulated influenza seasons:

- 2023/2024 (Moderate)
- 2024/2025 (High severity)
- 2025/2026 (Moderate-High)

Files include:

- National weekly influenza cases, hospitalisations and deaths.
- State-level weekly surveillance data for all 36 states and FCT.
- Weekly incidence per 100,000 population.
- Seasonal summaries (attack rate, peak week, peak incidence, cumulative burden, CFR).
- State metadata (population, geopolitical zone, climate classification, healthcare accessibility index).

Primary Task

- Forecast weekly influenza cases for Nigeria.

Secondary Tasks

- Forecast weekly hospitalisations.
- Forecast weekly deaths.

Seasonal Targets

- Peak week.
- Peak incidence.
- Seasonal cumulative cases.
- Seasonal cumulative hospitalisations.
- Seasonal cumulative deaths.
- Seasonal attack rate.

Spatial Targets

- Northern Nigeria.
- Southern Nigeria.
- All 36 states and FCT.

Probabilistic Targets

- Median forecast.
- 50% prediction interval.
- 90% prediction interval.
- Quantile forecasts.

Optional

- Probability of increasing transmission.
- Probability of decreasing transmission.
- Probability that epidemic peak occurs within specified weeks.

Dataset Characteristics

The synthetic surveillance dataset reproduces realistic influenza epidemiology including:

- Wet-season transmission.
- Harmattan-associated secondary waves.
- Climate-driven regional variation.
- State-level heterogeneity.
- Healthcare accessibility effects.
- Observation/reporting noise.
- Inter-season variability.

Evaluation Criteria

Forecasts will be evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)

- Weighted Interval Score (WIS)
- Continuous Ranked Probability Score (CRPS)
- Quantile Loss
- Peak-week accuracy
- Peak-incidence accuracy
- Seasonal cumulative burden accuracy
- National, regional and state-level forecasting performance
- Calibration of probabilistic forecasts

Submission Requirements

Participants shall submit:

1. Technical Report (single PDF)
2. Forecast dataset (CSV)
3. Source code
4. Documentation

The report should contain:

- Title
- Abstract
- Introduction
- Assumptions
- Forecasting methodology
- Model architecture
- Model justification
- Results
- Policy Implications sections
- Discussion
- Uncertainty analysis
- Conclusions
- References

Recommended Visualisations

- National weekly cases, hospitalisations and deaths.
- Regional epidemic curves.
- North versus South wave comparison.
- State-level weekly heatmaps.
- State attack-rate maps.
- State mortality maps.
- Peak-week summaries.
- Forecast uncertainty bands.

Rules

- Only the supplied simulated dataset may be used for model development and evaluation.
- Real influenza surveillance datasets must not replace the supplied synthetic data.
- Models must be original, reproducible, and scientifically justified.
- External libraries are permitted provided they are cited.
- Ensemble and hybrid approaches are allowed.

Grand Challenge

Develop an innovative, interpretable, and reproducible influenza forecasting framework capable of accurately predicting seasonal influenza dynamics across Nigeria using synthetic surveillance data while providing reliable probabilistic forecasts and actionable public health insights.

Submission Guidelines

- **Duration:** Saturday, 5th September 2026 (Problem Release) – Sunday, 5th October 2026 (Midnight).
- **Submission Deadline:** Participating teams and individuals are required to submit their completed work by **5th October 2026, before Midnight (West African Time / WAT)**.
- **Format:** All submissions must be compiled into a **single PDF document** including all required sections as specified in the competition guidelines. All accompanying code must be uploaded to **GitHub** and the repository link included within the submitted PDF.
- **Submission Portal:** Submissions must be uploaded exclusively through the official MAEPiMS 2.0 submission portal. Details will be shared with confirmed registered participants upon problem release on 5th September 2026.
- **Eligibility:** Only participants who have successfully completed their Expression of Interest (EOI) registration by **31st August 2026** are eligible to receive the competition problem and submit entries.
- **Submissions should align with epidemiological weeks (Sunday to Saturday)**

Contact Information

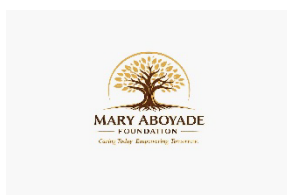
For enquiries regarding the **Nigeria Influenza Forecasting Challenge** MAEPiMS 2.0 Challenge, please contact:

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Participants will be provided with the competition dataset upon problem release on 5th September 2026. All data required for the challenge will be supplied directly to confirmed registered participants through the official MAEPiMS 2.0 portal. Participants are expected to use the supplied dataset to derive initial values, parameterize their models, and develop their analytical solutions.

This collaborative effort aims to enhance the understanding of infectious disease dynamics across Africa and inform data-driven decision-making that contributes to public health policy and pandemic preparedness on the continent.



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